
When Does Action-Type Supervision Help GUI Grounding?

Separating Real Effects from Evaluation Artifacts

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Abstract

GUI agents must predict an action type (*click*, *scroll*, *type*, ...) and a spatial target from a screenshot and an instruction. We study whether explicitly factoring these decisions helps a 2B-parameter agent, via a matched-compute comparison of five conditioning mechanisms on Qwen2-VL: a flat baseline (A), an auxiliary action-type loss (B), hard routing (C), an additive embedding hook (D-hook), and the prepended learned action embedding our original proposal advocated (D-token). Our first evaluation of this question was wrong in three independent ways, each of which we diagnose, quantify, and repair: (i) injecting conditioning via `inputs_embeds` silently bypasses Qwen2-VL’s M-RoPE position path and manufactured an 8σ *false negative*; (ii) train/validation splits taken as consecutive slices of one episode stream leaked shared task goals (24% of validation steps at the headline setting, up to 85% at scale) while looking clean at the episode level (2%); and (iii) a paired bootstrap pooling (seed, example) predictions as independent units was anti-conservative, rejecting a true clustered null 21.2% of the time at nominal 5% (an episode-clustered replacement achieves 5.1%). We then rebuilt the data pipeline around frozen goal-disjoint splits with a test region no phase of the project had ever touched, prespecified four contrasts, and evaluated once. **Every mechanistic conclusion of the original study reversed**, and the statistical repair alone would not have caught most of them. The auxiliary loss that the repaired statistics still endorsed on historical data (B–A +0.051 [+0.020, +0.081], $p = 0.0043$, versus the invalid pooled $p = 0.0002$) is null on the untouched test set (+0.011 [-0.005, +0.029], Holm $p = 0.309$); the prepended action embedding our proposal advocated and the course-era version reported as *refuted* in fact beats it (D-token–B +0.035 [+0.017, +0.054], Holm $p = 0.0027$); the gain is proportionally largest on *scroll* ($2.5\times$) rather than *click*; the control condition designed to show no effect shows one, and is therefore invalid; and the deployable predicted-type pipeline is null (+0.015 [-0.004, +0.034], Holm $p = 0.309$) because Stage 1 accuracy falls from 0.795 on validation to 0.708 on unseen goals, consuming the entire oracle gain. What survives is narrow and honest: *oracle* action-type conditioning improves grounding (D-hook–A +0.057 [+0.035, +0.079], Holm $p = 0.0004$), and nothing we can actually deploy does. We release the frozen splits, per-example predictions, and the clustered-inference code. A study can be multi-seed, ablated, controlled, and significant at $p < 0.002$ and still have all of its mechanistic claims invert under a clean split and a calibrated test.

1 Introduction

A GUI agent maps a screenshot and a natural-language instruction to a low-level action: a type (*click*, *scroll*, *type*, *drag*, ...) and, for spatial actions, a pixel-level target. State-of-the-art native agents (UI-TARS [13], OS-Atlas [11], SeeClick [4], ShowUI [7]) emit both in one autoregressive stream, entangling the two decisions. A documented failure mode follows: commit to the wrong action type, then ground *coherently* within the wrong semantic frame – scrolling past a visible target instead of clicking it. This motivated our original hypothesis, formulated as an architecture bet: a learned action-type embedding, prepended to the instruction, should let a grounding model condition on *what* before deciding *where*, and should help most in low data.

This paper reports what happened when we tested that bet carefully, and it has two intertwined subjects. The first is the scientific question: *when does action-type supervision change GUI grounding, and through which mechanism?* The second is methodological: *which implementation and evaluation choices can create a false appearance of benefit or harm in studies of exactly this kind?* We came to the second subject involuntarily – three separate artifacts in our own pipeline each produced a wrong conclusion at some point during the project – and we believe the post-mortem is at least as useful to practitioners as the object-level result.

Concretely, we compare five matched-compute Stage-2 conditioning mechanisms on Qwen2-VL-2B [10] with LoRA, over AITW [9] with a Mind2Web [5] cross-dataset control, preceded by a Stage-1 action-type classifier study. Our contributions:

- **A confirmatory answer, and it is mostly negative.** On an untouched, goal-disjoint test set: auxiliary action-type supervision does not measurably help; *oracle* inference-time conditioning does (+0.057 [+0.035, +0.079]); the deployable predicted-type pipeline does not, because the action-type classifier loses 0.795→0.708 accuracy on unseen task goals; and the proportionally largest effect is on *scroll*, not the click disambiguation our exploratory analysis reported.
- **Three quantified evaluation pitfalls.** (1) A silent M-RoPE / inputs_embeds interaction that fabricated an 8σ false negative; (2) episode-stream train/validation leakage that is invisible at the episode level (2% of validation steps) but large at the shared-task-goal level (24%–85%); (3) pseudo-replicated significance testing whose anti-conservativeness we measure on synthetic clustered nulls (21.2% type-I error at nominal 5%).
- **A repaired, releasable protocol.** Pinned-revision, goal-disjoint frozen train/validation/test manifests with an untouched test region beyond every historical data access; episode-clustered paired inference with per-seed effects and validated calibration; and a prespecified confirmatory analysis committed before test access. All artifacts, per-example predictions, and analysis code are released.
- **A measured cost of getting evaluation wrong.** Because we ran the same ablation twice — once under the flawed protocol, once under the repaired one — we can report what the flaws actually bought: not inflated *p*-values alone, but a reversal of which mechanism wins, which action class carries the gain, whether our own hypothesis was refuted, whether the pipeline is deployable, and whether the control holds (Section 7.4).

2 Related Work

End-to-end GUI agents. The dominant paradigm treats action type, arguments, and coordinates as one autoregressive output. UI-TARS [13] exemplifies this with a unified action space across platforms and explicit thought traces. OS-Atlas [11] scales the same paradigm with 13M cross-platform GUI grounding instances and shows that action-label harmonization across web/Android/desktop is nontrivial. ShowUI [7] builds a compact VLA around Qwen2-VL with interleaved vision-language-action streaming. Our flat baseline (Variant A) replicates the joint-decoding paradigm of these systems.

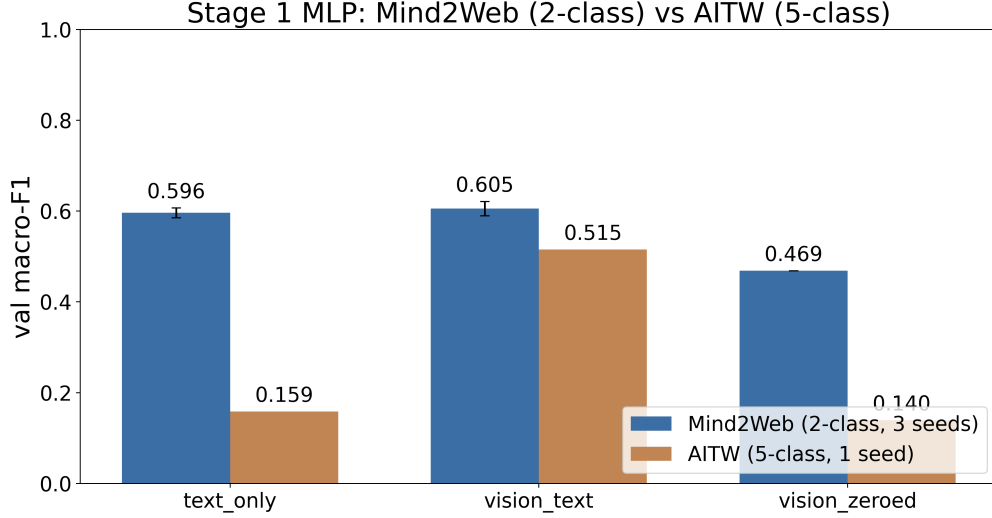


Figure 1: Stage 1 action-type classifier macro-F1 on Mind2Web (2 active classes) and AITW (5 active classes). Vision is essential on AITW (+0.414 over text-only) but redundant on Mind2Web (+0.009), because AITW instructions describe *goals* rather than physically specifying the action.

Modular grounding. SeeClick [4] argues that grounding is the central bottleneck in visual GUI agents. UGround / SeeAct-V [6] separates planning from grounding at the text-vs-pixel boundary. GUI-Actor [14] replaces coordinate-token regression with an attention-based action head driven by a dedicated <ACTOR> token plus a verifier. Our work studies a different module boundary: between *action-type prediction* and *spatial grounding*, not between planning text and coordinates.

Reward decomposition. UI-R1 [8] applies RL fine-tuning with a rule-based reward that separates action-type accuracy from action-argument accuracy on AndroidControl. It is the closest training analog: type and location as separable optimization targets; we use supervised conditioning rather than RL. RegionFocus [2] refines grounding at inference; we change the training-time conditioning pathway. SayCan [1] and ReAct [12] provide the modular-decision precedent, and RT-2 [3] the unified token-level action generation paradigm we critically examine.

Evaluation integrity. Our methodological findings echo, in a GUI setting, long-standing concerns about test-set contamination and dependence-blind inference: grouped data require grouped splits and cluster-aware uncertainty, and repeated measurements of the same example must not be pooled as independent evidence. Mind2Web’s own metric design (element accuracy vs. operation F1 vs. step success) [5] reflects the type/location decomposition we study.

3 Data and Splits

Sources. AITW [9] (HuggingFace mirror `cjfcst/AITW_General`, revision pinned to `5c0dc713`) is the primary Stage-2 benchmark; Mind2Web [5] provides the Stage-1 comparison and a cross-dataset control. Two AITW data mixes matter: `all_with_coords` (tap, swipe, type) and `taps_and_swipes`, intended as a control in which action type carries little information about *where* — an assumption Section 7.2 shows to be false. Targets are normalized to $[0, 1]^2$ and emitted as "`(x, y)`" on a $[0, 999]$ grid, with loss masked to coordinate tokens. Mirror decoding, the tap/swipe split rule, and class statistics are in Appendix A.

3.1 Two evaluation protocols

Exploratory protocol (Phases 1–7, historical). Our original runs streamed the AITW training split in mirror order, filtered by data mix, and took *consecutive slices*: the first n_{train} steps for training and the next n_{val} for validation. This is the kind of split that looks innocuous in code review and is not: episodes and, worse, repeated task goals straddle the boundary (Section 5.2). Every historical number in this paper is labeled exploratory and is re-analyzed with dependence-aware statistics.

Confirmatory protocol (Phase 9, frozen). We rebuilt the data pipeline around a frozen manifest: the first 12,000 `all_with_coords` steps of the pinned mirror revision, grouped by *normalized task goal* (episodes never span goals), split goal-disjointly into train (9,511 steps), validation (1,660), and an **untouched test region** (829): test groups are drawn exclusively from beyond stream index 5,500, past the furthest index any historical run ever loaded (5,250), so no test step was trained on, evaluated on, or inspected during the exploratory phase. Evaluation sets are frozen and identical for every variant and training size (250 validation / 600 test steps for `all_with_coords`; 200/400 for the control mix); training subsets are nested prefixes. The manifest, split hashes, 17 disclosed near-duplicate goal pairs (token Jaccard ≥ 0.8), duplicate checks, and the prespecified analysis (contrasts, metrics, corrections, failed-run rules) were committed to version control *before* any test-set evaluation ran. We claim “decisions frozen before test access,” documented by commit history – not third-party preregistration.

4 Methods

Stage 1. A frozen Qwen2-VL-2B encodes screenshot and instruction; mean-pooled final hidden states feed a 3-layer MLP trained with class-weighted cross-entropy (full details in Appendix A).

4.1 Stage 2: five conditioning mechanisms

Stage 2 fine-tunes Qwen2-VL-2B with LoRA (rank 16 on q/k/v/o projections, $\approx 0.2\%$ of parameters) to emit the coordinate string; all variants share data, optimizer ($\text{lr } 2 \times 10^{-5}$, 2 epochs, batch 1), and evaluation. The only difference is how action-type information enters:

A (flat baseline). No conditioning; replicates joint decoding.

B (auxiliary loss). A linear action-type head on the pooled representation adds cross-entropy to the grounding loss ($\lambda=1$); *no action information at inference*.

C (hard routing). Trained to emit “{word} (x, y)”; at inference the gold action word is forced as the decode prefix.

D-hook (additive embedding). A learned table $E \in \mathbb{R}^{8 \times d}$; the row for the current action type is added to *all* token embeddings via a forward hook, preserving the normal `input_ids` path (embedding norm at convergence ≈ 0.07 at the headline setting).

D-token (prepended token – the original hypothesis). A sentinel `<|action_slot|>` token is prepended to the instruction and its embedding replaced in place by $E[a]$, M-RoPE-correctly (the token is in `input_ids`); norm at convergence ≈ 2.2 , a strong routable signal.

All variants minimize the autoregressive loss over coordinate tokens, with variant-specific conditioning; B adds a weighted action-type cross-entropy term. The deployable end-to-end configuration trains D-hook, then a Stage 1 head on frozen-backbone features, and grounds under *predicted* types; gold-type results are oracle ceilings throughout (Appendix A).

4.2 Evaluation metrics

hit@ r : fraction of predictions within normalized L2 distance r of the target ($r \in \{0.05, 0.10, 0.25\}$; headline 0.10). **Mean L2:** mean normalized distance (parse failures scored at the $\sqrt{2}$ sentinel and counted as misses; parse rates reported). Statistical methodology is Section 5.3.

historical cell	val steps	sharing a train episode	sharing a train goal
all_wc, $n=300$	250	4 (2%)	14 (6%)
all_wc, $n=500$	250	4 (2%)	47 (19%)
all_wc, $n=800$	250	5 (2%)	49 (20%)
all_wc, $n=1200$	250	4 (2%)	61 (24%)
all_wc, $n=2500$	250	10 (4%)	133 (53%)
all_wc, $n=5000$	250	2 (1%)	213 (85%)
taps_swipes, $n=1000$	200	0 (0%)	48 (24%)

Table 1: Leakage in the historical (exploratory) splits: fraction of validation steps sharing an episode or a normalized task goal with that cell’s training slice. Goal sharing, not episode straddling, is the dominant channel.

5 Three Evaluation Pitfalls, Quantified

5.1 Pitfall 1: M-RoPE vs. `inputs_embeds`

Qwen2-VL assigns 3D (temporal, height, width) position IDs to visual tokens via an `input_ids`-keyed path. Our first D-token implementation injected the action embedding by passing `inputs_embeds` directly, which silently bypasses that path: image-token positions were computed without the action slot in the sequence, misaligning multimodal position embeddings for *every* visual token. The observed effect was a confident 8σ “A beats D” result – a pure artifact. Rebuilding the same mechanism through the `input_ids` path (D-hook, D-token) erased the deficit. Any conditioning study on M-RoPE-family models that touches `inputs_embeds` should verify position-ID computation explicitly; the artifact is silent, large, and directionally misleading.

5.2 Pitfall 2: stream-slice splits leak task goals

Table 1 quantifies the historical protocol’s leakage by reconstructing, for every historical cell, which episodes and normalized task goals each validation step shared with that cell’s training slice. Episode-level overlap – the leakage one usually checks – is small (2% at the headline cell). The dominant channel is *shared task goals*: 24% of headline validation steps have a goal string that also appears in training, rising to 85% at $n_{\text{train}}=5000$ – AITW goals repeat across many episodes, so consecutive-slice splitting almost guarantees template overlap at scale. Val-set difficulty therefore varies with n_{train} (each size had a different val slice), quietly confounding the historical scaling curve. The confirmatory protocol removes both channels by construction (goal-disjoint splits; Section 3.1).

5.3 Pitfall 3: pseudo-replicated significance

Our original headline significance pooled per-example distances across seeds – treating 3×250 (seed, example) rows as 750 independent paired observations. Predictions of the same example by different seeds are strongly correlated, and examples within an episode share state; the pooled bootstrap therefore overstates evidence. On synthetic nulls matching the data’s dependence structure (episode and example effects shared across seeds; zero true difference), the pooled test rejects 21.2% of the time at nominal 5% (1000 simulations), while the replacement below achieves 5.1%.

Repaired inference. For each contrast we (i) average each example’s paired difference over seeds, (ii) cluster examples by episode, (iii) bootstrap clusters (10,000 resamples) for 95% CIs, and (iv) use an episode-level sign-flip permutation test (10,000 permutations; p -values reported no smaller than Monte Carlo resolution – never $p=0$). **Estimand.** This tests the *fixed trained checkpoints* on new data from the eval population. It is *not* calibrated for claims about the training procedure: adding run-level (seed) offsets to the synthetic null drives the episode-clustered test to 55% rejection (and the pooled test to 70%). With only three seeds per variant we therefore report per-seed means everywhere and make no procedure-level significance claims. The retracted pooled p -values (including a headline $p=0.0002$) should not be cited.

variant	hit@0.10	mean L2 ↓	Δ hit@0.10 vs. A [95% CI]	perm. p (Holm)
A (flat)	0.219 ± 0.018	0.450	—	—
B (aux loss)	0.230 ± 0.004	0.445	+0.011 [-0.005, +0.029]	0.23 (0.309)
C (hard routing)	0.219 ± 0.019	0.498	+0.000 [-0.022, +0.021]	1
D-hook (additive)	0.276 ± 0.003	0.376	+0.057 [+0.035, +0.079]	$< 10^{-4}$ (0.0004)
D-token (prepended)	0.265 ± 0.004	0.388	+0.046 [+0.027, +0.067]	0.0002

Table 2: **Confirmatory** main ablation on the untouched, goal-disjoint AITW test set ($n=600$ steps; $n_{\text{train}}=1200$; 3 seeds; mean $\pm$ std across seeds). Δ vs. A: episode-clustered 95% CI and permutation p (Holm-adjusted across the four primary contrasts where applicable).

6 Exploratory Results (historical splits, repaired statistics)

All numbers in this section come from the leaky historical splits and are therefore *exploratory*; they are shown with episode-clustered statistics, and Section 7 gives the prespecified untouched-test versions of every primary claim.

The Stage 1 action-type classifier study (Table 7) is summarized by Figure 1 and given in full in Appendix B.

6.1 Main ablation under clustered statistics

Table 4 re-scores the historical headline with the repaired statistics (per-seed values in Figure 3). Three observations. First, the auxiliary-loss effect survives: B–A on hit@0.10 is +0.051 [+0.020, +0.081] ($p = 0.0043$), where the invalid pooled analysis had claimed $p = 0.0002$ – the conclusion stands but the evidence was overstated by an order of magnitude. Second, D-hook weakens to marginal after multiplicity correction. Third, the comparison our proposal cared about most – D-token vs. B – is *no longer significant* (-0.036 [-0.077, +0.005], $p = 0.114$): on the historical data we cannot even claim the prepended embedding is significantly *worse* than the auxiliary loss, only that it shows no sign of being better.

The exploratory per-class decomposition (Appendix B, Section B.2) attributed the advantage to *click*; Section 7.1 shows the confirmatory data do not.

The exploratory low-data grid (Appendix B) shows no clean monotone trend and is confounded by per-size validation sets.

Under clustered statistics the historical end-to-end “win” is already not significant (+0.016 [-0.017, +0.052], $p = 0.424$; Appendix B).

7 Confirmatory Results (untouched test, prespecified)

All variants were retrained from scratch on the frozen goal-disjoint splits (identical hyperparameters, seeds 42–44) and evaluated once on the untouched 600-step test set; the four primary contrasts, the headline metric (hit@0.10), the clustered test, and Holm correction were committed before any test evaluation ran (Section 3.1).

P1 (auxiliary supervision) — null. B vs. A on hit@0.10: +0.011 [-0.005, +0.029], permutation $p = 0.23$, Holm $p = 0.309$; mean L2 -0.005 [-0.016, +0.006] ($p = 0.42$). The variant the exploratory analysis identified as the winner does *not* replicate. Auxiliary action-type supervision, which adds no information at inference, leaves grounding statistically unchanged on data whose task goals were never seen in training.

P2 (inference-time conditioning) — positive, but oracle. D-hook vs. A: +0.057 [+0.035, +0.079] (Holm $p = 0.0004$; mean L2 -0.074 [-0.100, -0.050]), consistent in sign and size across all three seeds. This is the one large effect that survives the clean split. It is an *oracle* result: D-hook receives

variant	click ($n=393$)	scroll ($n=136$)	type ($n=71$)
A (flat)	0.312	0.064	0.000
B (aux loss)	0.337	0.042	0.000
C (hard routing)	0.310	0.113	0.000
D-hook (additive)	0.366	0.162	0.000
D-token (prepended)	0.367	0.108	0.000

Table 3: Confirmatory per-class hit@0.10 on the untouched test set (seed-averaged). The oracle conditioning gain is proportionally largest on *scroll* (A 0.064 $\rightarrow$ D-hook 0.162, a $2.5\times$ relative improvement) and modest on *click* (0.312 $\rightarrow$ 0.366). The auxiliary loss (B) improves click slightly while *harming* scroll. *type* remains exactly 0.000 for every variant: AITW type coordinates are degenerate.

the ground-truth action type at test time, which A never sees. P4 is what happens when that oracle is replaced by a real classifier.

P3 (the proposal’s mechanism) — reversed in our favour, and we flag it as such. D-token vs. B: $+0.035$ [$+0.017, +0.054$] (Holm $p = 0.0027$) — the prepended learned action embedding our original proposal advocated significantly *outperforms* the auxiliary loss on the clean test set. The exploratory data pointed the other way (Section 6), and the course-era version of this work reported the embedding as refuted. On the untouched test both embedding mechanisms beat B (D-token vs. A $+0.046$ [$+0.027, +0.067$], $p = 0.0002$; D-hook vs. B $+0.046$ [$+0.026, +0.066$], $p = <10^{-4}$), while hard routing is exactly null (C vs. A $+0.000$ [$-0.022, +0.021$], $p = 1$). We stress that this contrast reverses in the direction that flatters our own hypothesis, which is precisely why it needed a prespecified test on data we had never touched.

P4 (deployable pipeline) — null. End-to-end predicted-type pipeline vs. A: $+0.015$ [$-0.004, +0.034$] (Holm $p = 0.309$). The reason is Stage 1 generalization, not Stage 2: the action-type classifier scores 0.795 accuracy on the frozen validation split but only 0.708 on the untouched test split, and the paired predicted-vs-oracle gap is -0.048 [$-0.061, -0.035$] ($p = <10^{-4}$) — larger than the entire oracle advantage over A. **Classifier error consumes the whole benefit**, directly contradicting the historical claim that “classifier error is not the bottleneck.”

7.1 Where the confirmatory gain lives: scroll, not click

Table 3 overturns the exploratory mechanism story of Section B.2. On leaky data the conditioning advantage looked like *click* disambiguation. On the untouched test set the dominant relative effect is on *scroll* ($+0.098$ absolute, $2.5\times$ relative), with a smaller click gain ($+0.053$). The mechanism this supports is different and simpler: swipe targets occupy a narrow, largely goal-independent region of the screen, so being told “this step is a scroll” relocates the prediction in a way the flat baseline cannot infer from the screenshot and instruction alone. B, which gets the same supervision only as a training signal, captures none of it and slightly harms scroll.

7.2 The control condition does not behave as designed

Our preregistered control assumed that on *taps_and_swipes* action type carries no spatial information, so conditioning should not help. It does help: D-hook vs. A on the control test set is $+0.047$ [$+0.010, +0.079$] ($p = 0.0327$), while B is null ($+0.013$ [$-0.013, +0.041$], $p = 0.364$) and C is significantly harmful (-0.067 [$-0.100, -0.039$], $p = <10^{-4}$). We report this as a **failed control**, not a second confirmation. It is coherent with Section 7.1: if the exploitable signal is “swipes land in a predictable region,” then a tap-versus-swipe mix is exactly a setting where action type *is* spatially informative, and our control was mis-designed rather than uninformative. The consequence for the paper is concrete: we cannot claim “conditioning helps if and only if action type is spatially predictive,” because we no longer have a matched condition in which it demonstrably is not.

7.3 Low data

The confirmatory low-data grid (Table 6) evaluates every training size against the *same* frozen test set, removing the per-size-val-set confound that made Figure 4 unreadable. D-hook’s oracle advantage is present at $n=300$ and $n=800$ and not resolved at $n=500$; B is positive only at $n=300$. There is no clean monotone low-data trend in either variant, so we again decline the “low-data prior” claim our proposal predicted.

Secondary contrasts, per-class decompositions, parse rates, and per-seed values for every run are in the released `results/phase9_rerun/rerun_analysis.json`. No procedure-level significance claim is made from three seeds (Section 5.3); per-seed means accompany every contrast.

7.4 What the broken evaluation actually cost

Because the same five-variant ablation was run under both protocols, we can separate the two defects and ask what each one cost. Three states are distinguishable: **(i)** the original analysis (leaky split, pooled bootstrap) — what a course-era version of this paper reported; **(ii)** the statistically repaired analysis on the *same* leaky data (Section 6); and **(iii)** the confirmatory analysis on the untouched split.

- **Which variant wins.** (i) B beats A, $p=0.0002$. (ii) B still beats A, $+0.051$ [$+0.020$, $+0.081$], $p=0.0043$. (iii) *B is null* ($+0.011$ [-0.005 , $+0.029$]); the embedding mechanisms are the only ones that help.
- **Whether our own hypothesis was refuted.** (i) D-token is significantly worse than B (pooled $p=0.0192$), so the proposal’s prepended embedding is refuted. (ii) the gap is no longer significant (-0.036 [-0.077 , $+0.005$], $p=0.114$) — the refutation was already an artifact of the pooled test. (iii) *D-token significantly beats B* ($+0.035$ [$+0.017$, $+0.054$]).
- **Where the gain lives.** (i)–(ii) click disambiguation. (iii) proportionally, *scroll* ($2.5\times$ versus $1.2\times$ for click).
- **Whether the pipeline is deployable.** (i) the predicted-type pipeline beats A and “classifier error is not the bottleneck.” (ii) the pipeline gain is no longer significant ($+0.016$ [-0.017 , $+0.052$]). (iii) the pipeline is *null* and classifier error consumes the entire oracle gain.
- **Whether the control holds.** (i)–(ii) conditioning looks neutral where action type was assumed uninformative. (iii) the control shows a significant effect, so it is *not a valid control*.

Two lessons separate cleanly. First, the statistical repair alone was *not* sufficient: it corrected two claims (the D-token refutation and the deployable pipeline) but left the headline “auxiliary supervision works” and the click-mechanism story intact and wrong — those required the clean split. Fixing inference without fixing the data would have produced a paper that was better calibrated and still substantively mistaken. Second, the direction of error is not systematic: the flawed protocol inflated some effects, suppressed our own hypothesis, and mis-located the mechanism. A study can be internally consistent, multi-seed, ablated, controlled, and significant at $p < 0.002$, and still have every one of its mechanistic claims move under a clean split and a calibrated test.

8 Limitations

The single positive confirmatory effect is oracle-gated: D-hook and D-token receive ground-truth action types that a deployed agent would have to predict, and when we predict them the effect disappears (P4). No claim in this paper should be read as a deployable improvement. Our control condition failed (Section 7.2), so we have no valid matched setting in which action type is demonstrably uninformative; the scope condition on when conditioning helps is therefore untested, and designing a genuine control (for instance, targets randomized within class) is future work. The

300 scroll-versus-click mechanism reading in Section 7.1 is a post-hoc decomposition of confirmatory
301 data, not a prespecified test.

302 Offline step-level hit@radius is not long-horizon task success: it scores logged single steps, without
303 interaction, recovery, or compounding-error dynamics – the motivating failure mode is measured
304 here only via its step-level signature. The backbone is a single 2B model; conclusions are about
305 this scale and training recipe, not about GUI agents generally, and our baselines are small matched-
306 compute models trained by us, not state-of-the-art systems. The confirmatory estimand is fixed
307 checkpoints, not the training procedure (three seeds; per-seed effects reported). AITW’s type coord-
308 inates are degenerate, so “grounding” on `all_with_coords` is effectively click/scroll grounding
309 plus a constant; the click-concentration analysis is post-hoc on exploratory data and only its prespec-
310 ified aggregate versions are confirmatory. Goal strings are our dependence proxy; 17 near-duplicate
311 train/test goal pairs (disclosed in the manifest) survive normalization, and subtler sharing (same app,
312 similar screens) is not modeled. The Mind2Web control used 2 seeds without per-example logs. Fi-
313 nally, the mirror we pin is a third-party repackaging of AITW; we verified revision stability but not
314 byte-level provenance against the original release.

315 9 Conclusion

316 On a test set whose task goals were never seen in training, the factorization thesis survives only
317 in oracle form. Telling a 2B GUI agent the ground-truth action type improves grounding (+0.057
318 [+0.035, +0.079] hit@0.10), and the prepended action embedding our proposal advocated is the
319 better of the two mechanisms that do this (D-token-B +0.035 [+0.017, +0.054]) — a reversal of the
320 course-era conclusion that the embedding was refuted. But auxiliary supervision alone is null, hard
321 routing is null, the effect concentrates in *scroll* rather than the click disambiguation we previously
322 reported, our control condition turns out not to be a control, and the only configuration one could
323 actually deploy — predicting the action type and conditioning on the prediction — is statistically
324 indistinguishable from doing nothing, because the classifier loses 0.795→0.708 accuracy on unseen
325 goals. The honest summary is that we measured a real but oracle-gated effect and no deployable
326 one.

327 The methodological result is the durable one. We ran the same five-variant ablation twice: once un-
328 der a protocol with three defects (an `inputs_embeds/M-RoPE` interaction fabricating an 8σ false
329 negative, goal-level split leakage reaching 85%, and a pooled bootstrap with 21.2% measured type-I
330 error), and once under a repaired one. The defects did not merely exaggerate significance; they
331 inverted which mechanism wins, which class carries the gain, whether our hypothesis was refuted,
332 whether the pipeline was deployable, and whether the control held. None of the three is exotic
333 — stream-sliced splits, embedding-injection shortcuts, and pooled per-seed bootstraps are ordinary
334 practice — so we expect the same reversals are latent in other offline GUI-grounding evaluations.
335 The frozen splits, per-example predictions, clustered-inference code, and prespecification are re-
336 leased so that both the object-level result and the audit can be checked and rerun.

337 Contributions

338 **A. Chauhan:** training and evaluation infrastructure, Stage-2 conditioning variants, cloud experi-
339 ment orchestration, statistical reanalysis and confirmatory protocol, writing. **A. Ilyasov:** Stage-1
340 classifier study, data pipeline and error analysis, figures, editing. Both authors contributed to prob-
341 lem formulation and experimental design and approved the final manuscript.

342 Generative AI Usage Statement

343 Large-language-model assistants (Anthropic Claude) were used extensively under the authors’ direc-
344 tion: writing and debugging training, evaluation, and statistical-analysis code; orchestrating cloud
345 experiments; drafting the integrity audit that led to Sections 5 and 7; and drafting and editing this

manuscript. The authors specified the research questions and methodology, reviewed the analyses, verified reported numbers against the committed result artifacts (every table is generated programmatically from those artifacts), and take full responsibility for all content.

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A Method details

A.1 Sources and preprocessing

We use two datasets under a canonical 8-class action taxonomy (*click*, *double_click*, *type*, *scroll*, *drag*, *hotkey*, *wait*, *finished*).

AITW [9] (via the `cjfcst/AITW_General` HuggingFace mirror, revision pinned to `5c0dc713`) is the primary Stage-2 benchmark: five active classes (click 55%, scroll 13%, finished 12%, type 11%, hotkey 9%). Screenshots in the mirror are raw RGB bytes without headers; we decode them against seven observed resolutions with a portrait-aspect fallback. `DUAL_POINT` events are split into taps (click) versus swipes (scroll) by normalized touch–lift distance (threshold 0.04). Two data mixes matter below: `all_with_coords` (tap, swipe, type – action type is spatially predictive because taps target small elements while swipes are gestures) and `taps_and_swipes` (a control where knowing the type carries little information about *where*).

Mind2Web [5] (multimodal release) provides the Stage-1 cross-dataset comparison and a Stage-2 cross-dataset control; it is click/type-dominated with only two active classes at scale.

Coordinates. Targets are normalized to $[0, 1]^2$ and rendered as the string "`(x, y)`" on a $[0, 999]$ grid; training loss is masked to the coordinate tokens.

variant	hit@0.10	mean L2 ↓	Δ hit@0.10 vs. A [95% CI]	perm. p
A (flat)	0.255 ± 0.021	0.392	—	—
B (aux loss)	0.305 ± 0.012	0.362	+0.051 [+0.020, +0.081]	0.0043
C (hard routing)	0.279 ± 0.032	0.412	+0.024 [-0.016, +0.063]	0.278
D-hook (additive)	0.300 ± 0.028	0.365	+0.045 [+0.011, +0.085]	0.0227
D-token (prepended)	0.269 ± 0.025	0.375	+0.015 [-0.018, +0.047]	0.437

Table 4: Exploratory main ablation (AITW `all_with_coords`, $n_{\text{train}}=1200$, 3 seeds, historical leaky splits). Episode-clustered Δ vs. A with 95% cluster-bootstrap CIs and sign-flip permutation p . Under the repaired analysis, B survives (Holm-adjusted $p = 0.017$ across the four vs.-A contrasts); D-hook is marginal after correction (Holm $p = 0.068$); C and D-token are not significant.

variant	hit@0.10	mean L2 ↓	Δ hit@0.10 vs. A [95% CI]	perm. p
A (flat)	0.302 ± 0.007	0.268	—	—
B (aux loss)	0.316 ± 0.014	0.264	+0.013 [-0.013, +0.041]	0.364
C (hard routing)	0.236 ± 0.052	0.454	-0.067 [-0.100, -0.039]	$< 10^{-4}$
D-hook (additive)	0.349 ± 0.016	0.213	+0.047 [+0.010, +0.079]	0.0327

Table 5: Confirmatory control mix (`taps_and_swipes`) on its frozen 400-step test set. The control was designed as the falsifiable half of the thesis — conditioning should not help where action type is uninformative — and it *fails*: D-hook shows a significant gain here too (Section 7.2).

393 A.2 Stage 1: action-type classifier

394 A frozen Qwen2-VL-2B encodes screenshot+instruction; mean-pooled final hidden states (1536-
395 d) feed a 3-layer MLP trained with class-weighted cross-entropy (AdamW, lr 10^{-4} , batch 128, 8
396 epochs, best epoch by validation macro-F1). Modes: `text_only` vs. `vision_text`.

397 **Objective.** With screenshot s , instruction g , action type a , coordinate tokens $y_{1:T}$ and variant-
398 specific conditioning c :

$$\mathcal{L}_{\text{coord}} = - \sum_{t=1}^T \log p_{\theta}(y_t \mid y_{<t}, s, g, c), \quad (1)$$

399 with $c = \emptyset$ for A and B, $c = E[a]$ for D-hook/D-token, and the forced word for C; B adds
400 $\lambda \text{CE}(h_{\phi}(\bar{z}), a)$ (Eq. 1 otherwise unchanged).

401 **End-to-end pipeline.** The deployable configuration trains D-hook, then a small MLP head on
402 frozen-backbone features (Stage 1 in-job), and evaluates grounding under *predicted* types; gold-
403 type results are oracle ceilings, never deployable claims.

404 B Exploratory material (historical splits)

405 This appendix holds supporting exploratory material referenced from the main text: the Stage 1
406 action-type classifier study, the D-token causal-use sensitivity analysis, and three figures from the
407 historical (leaky-split) phase. All of it is exploratory and superseded for inferential purposes by the
408 confirmatory results in the main text.

409 B.1 Stage 1: vision is essential exactly where instructions under-specify the action

410 Table 7 and Figure 1: on Mind2Web, instructions name the action (“click the submit button”), so
411 vision adds +0.009 macro-F1 over text; on AITW, instructions state goals (“open a new Chrome
412 private window”), text-only collapses toward the majority class, and vision adds +0.414. A TF-IDF
413 ceiling of 0.168 on AITW confirms text is uninformative there regardless of classifier strength. This
414 asymmetry decides where Stage-2 conditioning could matter at all.

train size	A hit@0.10	Δ B–A [95% CI]	Δ D-hook–A [95% CI]
$n=300$	0.138	+0.038 [+0.013, +0.064]	+0.049 [+0.025, +0.073]
$n=500$	0.219	−0.004 [−0.027, +0.018]	+0.016 [−0.008, +0.039]
$n=800$	0.256	+0.003 [−0.014, +0.020]	+0.036 [+0.015, +0.057]

Table 6: Confirmatory low-data grid on the *same* frozen test set for every training size (nested training prefixes), removing the per-size-val-set confound of Figure 4.

Variant	Mind2Web macro-F1	AITW macro-F1
text_only	0.597 ± 0.011	0.141 ± 0.032
vision_text	0.605 ± 0.016	0.555 ± 0.036
vision_zeroed	0.469 ± 0.000	0.110 ± 0.051
Vision delta	+0.009	+0.414
TF-IDF + LogReg	0.622	0.168
Majority floor	0.472	0.140

Table 7: Stage 1 action-type classification (3 seeds, mean $\pm$ std; exploratory splits). On Mind2Web the instruction text alone contains the action type; on AITW it does not, and vision contributes +0.414 macro-F1. `vision_zeroed` collapsing to the majority floor is the sanity check that features, not priors, drive the result.

415 B.2 Per-class decomposition: the gain is click disambiguation

416 Breaking hit@0.10 down by class (Figure 2): B gains +0.120 on click (68% of examples) with a
417 small scroll cost; C has the largest click gain (+0.122) but destroys scroll (−0.144) by forcing a de-
418 code prefix that derails the gesture class; D-hook alone improves both (click +0.058, scroll +0.039).
419 The *type* class scores exactly 0.000 across all variants with mean normalized L2 at or slightly above
420 $\sqrt{2}$, the corner-to-corner distance (a few predictions land outside the unit square): AITW *type*
421 events carry degenerate touch coordinates and are ungroundable by any model. This corrects our
422 earlier “type→text-field localization” interpretation: the real signal is **click-vs-scroll/type disam-**
423 **biguation** for the precise-target class, and the “type” rows merely add a constant drag that cancels
424 in deltas.

425 B.3 Control conditions

426 On `taps_and_swipes` (action type spatially uninformative), the historical 3-seed means show no
427 conditioned variant meaningfully above A (A 0.390 ± 0.008 , D-hook 0.392 ± 0.035 – a difference
428 well inside one seed-std – with B 0.368 ± 0.009 and C 0.325 ± 0.032 at or below A on hit@0.10);
429 per-example logs were not retained for A on this mix, so the clustered version of this control appears
430 only in the confirmatory rerun (Table 5). On Multimodal-Mind2Web (click/type-dominated), D-
431 hook vs. A was −0.006 hit@0.10 / +0.018 hit@0.25 over 2 seeds – a null, retained here deliberately:
432 conditioning does not help where action type carries no spatial information, on either dataset.

433 B.4 Low-data behavior: no clean curve

434 Our proposal predicted the advantage is a low-data prior. The historical grid ($n \in \{300, 500, 800\}$,
435 3 seeds) does *not* show a clean monotone: B–A on hit@0.10 is +0.067 at $n=300$ but +0.011 at
436 $n=500$ and −0.040 at $n=800$; D-hook is more stable (+0.069, +0.001, +0.048) and leads hit@0.25
437 and mean L2 at every low-data size. The two single-seed high-end cells disagree with each other:
438 at $n=5000$ the deltas shrink and partially reverse (B–A hit@0.25 goes negative), while at $n=2500$
439 D-hook posts the *largest* conditioning deltas anywhere in the study (+0.076 hit@0.10, +0.204
440 hit@0.25) – single-seed noise on a differently-leaky val slice, cutting in the direction *favorable*
441 to conditioning. We report all of this as: *inconclusive on the low-data-prior hypothesis – noisy*,

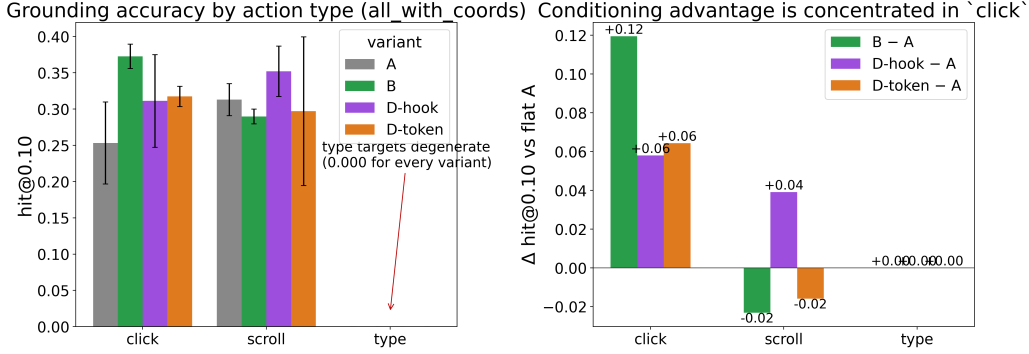


Figure 2: Exploratory hit@0.10 by action class (all_with_coords). The conditioning advantage concentrates in *click*; *type* is degenerate on AITW (all variants score exactly 0.000, mean L2 at the $\approx \sqrt{2}$ corner-to-corner distance).

contrast	Δ hit@0.10 [95% CI]	Δ hit@0.25 [95% CI]	perm. p (hit@0.10)
wrong – gold	-0.192 [-0.233, -0.149]	-0.329 [-0.372, -0.284]	$< 10^{-4}$
zero – gold	-0.093 [-0.128, -0.059]	-0.112 [-0.145, -0.079]	$< 10^{-4}$

Table 8: Exploratory D-token sensitivity (episode-clustered, 3 seeds): evaluating the same trained model with wrong or zeroed action embeddings severely degrades grounding. This establishes the embedding is *used* at inference – a sensitivity/oracle analysis, not evidence that conditioning caused any gain over A.

442 *non-monotone, and confounded by per-size val sets* (Figure 4); the prespecified low-data claim is
 443 evaluated on the frozen test set in Section 7. An earlier draft of this work called the low-data
 444 prediction “confirmed”; that claim did not survive this audit.

445 B.5 The e2e pipeline claim also weakens

446 The historical end-to-end result (predicted-type pipeline 0.271 vs. A 0.255 hit@0.10) was previ-
 447 ously reported as a win. Under clustered statistics the paired difference is +0.016 [-0.017, +0.052]
 448 ($p = 0.424$) – not significant; the oracle-type version is marginal (+0.037 [+0.005, +0.076]). The
 449 deployable-pipeline question is therefore decided by the confirmatory run (P4, below), not the his-
 450 torical data.

451 A sensitivity analysis feeding the trained D-token model wrong or zeroed action embeddings (Ap-
 452 pendix B, Table 8) confirms the embedding is causally used at inference rather than ignored; it is a
 453 sensitivity test, not evidence that conditioning caused any gain over the baseline.

454 B.6 D-token causal-use sensitivity

455 Feeding the trained D-token model a *wrong* action id collapses grounding, and zeroing the embed-
 456 ding hurts substantially (Table 8). The refutation of the proposal’s mechanism is therefore precise:
 457 the prepended embedding is causally used, it is simply not the best way to exploit action-type super-
 458 vision. Figure 5 shows one example where correct conditioning concentrates more attention mass
 459 on the screenshot; across the three examples we probed the pattern is not uniform, so we present it
 460 strictly as illustration, not evidence.

Stage 2 ablation (AITW all_with_coords): $B \approx D\text{-hook} > C > A$

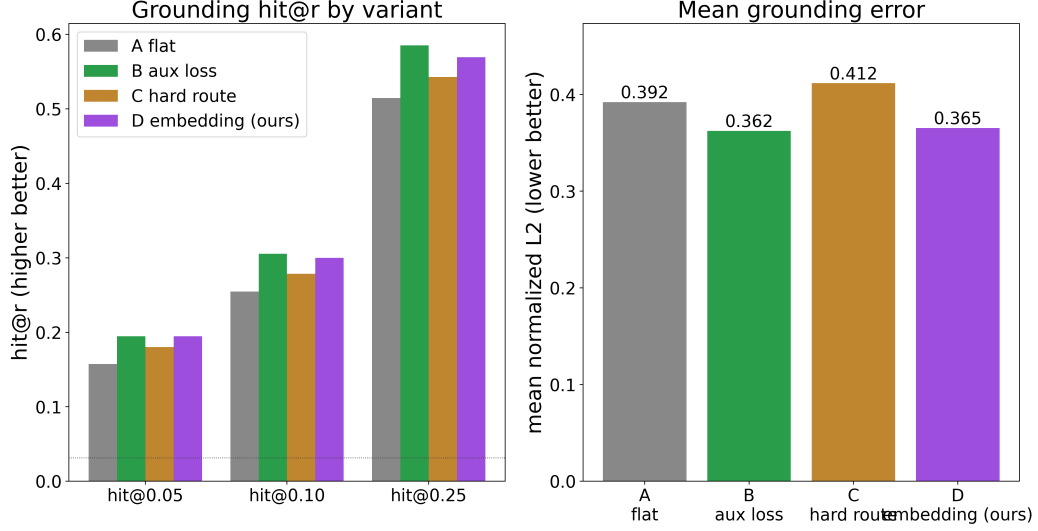


Figure 3: Exploratory ablation on all_with_coords (historical splits; per-seed values). Shown for continuity with the historical record; Table 4 carries the calibrated statistics and Table 2 the confirmatory test-set versions.

Data-scaling: action-type conditioning advantage vs training size
(AITW all_with_coords; 300-1200 are 3-seed means, 2500/5000 single seed)

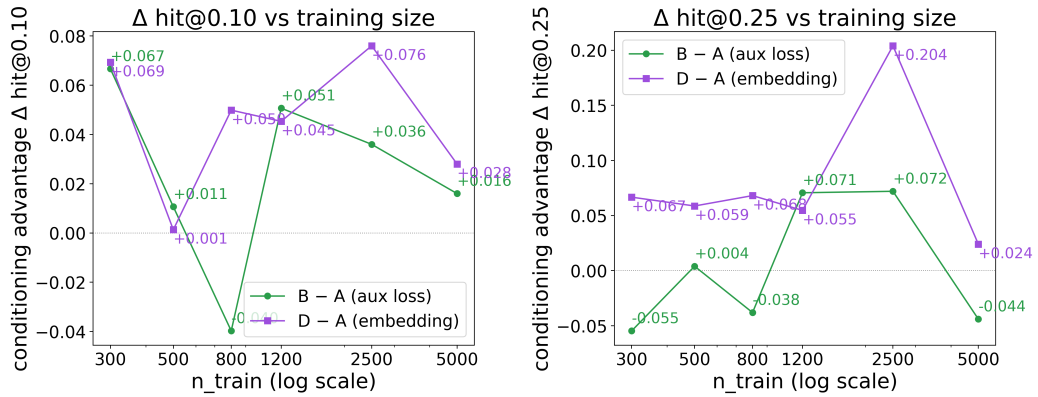


Figure 4: Exploratory conditioning advantage ($\Delta \text{hit@0.10}$ vs. A) across n_{train} . Caveat from Section 5.2: each historical size used a *different* val slice with different goal leakage (up to 85%), so this curve is directional at best; the confirmatory grid (Table 6) fixes one frozen test set for all sizes.



Figure 5: Image-region attention for a held-out *click* example under three action-type conditionings (qualitative illustration; screenshot attention mass 0.204 under *click* vs. 0.132/0.110 under *typelscroll*).